Trøndelagkonkurranse 2026 - Level II

Third Round - November 3, 2026

1st Part

Choose and solve only one of the following problems:

Problem 1

Let $A = \{1, 3, 4\}$. We can take any two different elements of A, say a and b, and replace them with c and d such that

$$c = \frac{3a - 4b}{5} \quad \text{and} \quad d = \frac{4a + 3b}{5}$$

Let $B = \{2, 3, 5\}$. Determine whether A can be transformed into B through successive transformations.

Problem 2

A natural number $n > 1$ is called *agreeable* if an equilateral triangle T that can be divided into n smaller equilateral triangles such that:

- The sides of the smaller triangles are parallel to the sides of T.
- The smaller triangles do not overlap inside T. Vertices and edges can touch each other.
- Every portion of T is covered by a smaller triangle.

Show that 2026 is agreeable.

Problem 1

Let's take $a, b \in A$. Observe first that

$$c^2+d^2 = \left(\frac{3a-4b}{5}\right)^2+\left(\frac{4a+3b}{5}\right)^2 = \frac{9a^2-24ab+16b^2}{5^2}+\frac{16a^2+24ab+9b^2}{5^2} = \frac{25a^2+25b^2}{5^2} = a^2+b^2,$$

which means that applying this process preserves the value of adding up the squares of the elements of A. Then, since for the elements in A we have

$$1^2+3^2+4^2 = 1+9+16 = 26$$

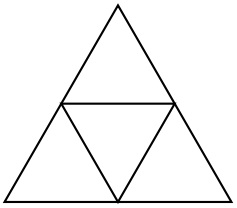
and for the elements in B we have

$$2^2+3^2+5^2 = 4+9+25 = 38,$$

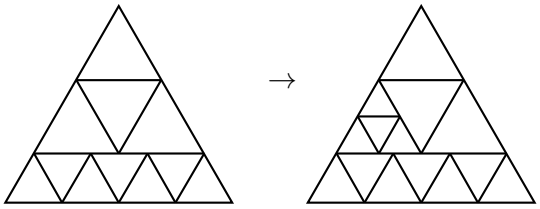
A can never be transformed into B.

Problem 2

We will justify that all the numbers n of the form $n = 3k+1$ are agreeable. First, $n = 4$ is agreeable by one of the most common ways of dividing T



Suppose now that a certain value n is agreeable, then, taking one of its triangles and dividing it as above, we have increased the number of triangles by 3. Here is an example from 11 to 14.



Thus, starting from T, we can always add 3 equilateral triangles following the same procedure. This means that the values

$$1, 4, 7, 10, 13, 16, \ldots, 3k+1, \ldots$$

are agreeable. Finally, since

$$2026 = 3 \cdot 675 + 1$$

it follows that $n = 2026$ is agreeable.